IMPLEMENTATION OF DIFFERENT NETWORK TOPOLOGIES USING CISCO PACKET TRACER

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AIM

To implement and simulate Star, Bus, and Mesh network topologies using Cisco Packet Tracer and verify their working using ping command and simulation mode.

SOFTWARE REQUIRED

• Cisco Packet Tracer

THEORY

Network topology refers to the physical or logical arrangement of devices in a network.

STAR TOPOLOGY:
All devices are connected to a central switch. Communication happens through the switch.
Advantages: Easy management, easy troubleshooting, better performance.
Disadvantages: If switch fails, entire network fails.

BUS TOPOLOGY:
All devices are connected to a single backbone cable.
Advantages: Low cost, less cable required.
Disadvantages: Backbone failure stops entire network.

MESH TOPOLOGY:
Each device is connected to every other device.
Advantages: High reliability, fault tolerant.
Disadvantages: Expensive and complex installation.

IP ADDRESSING USED

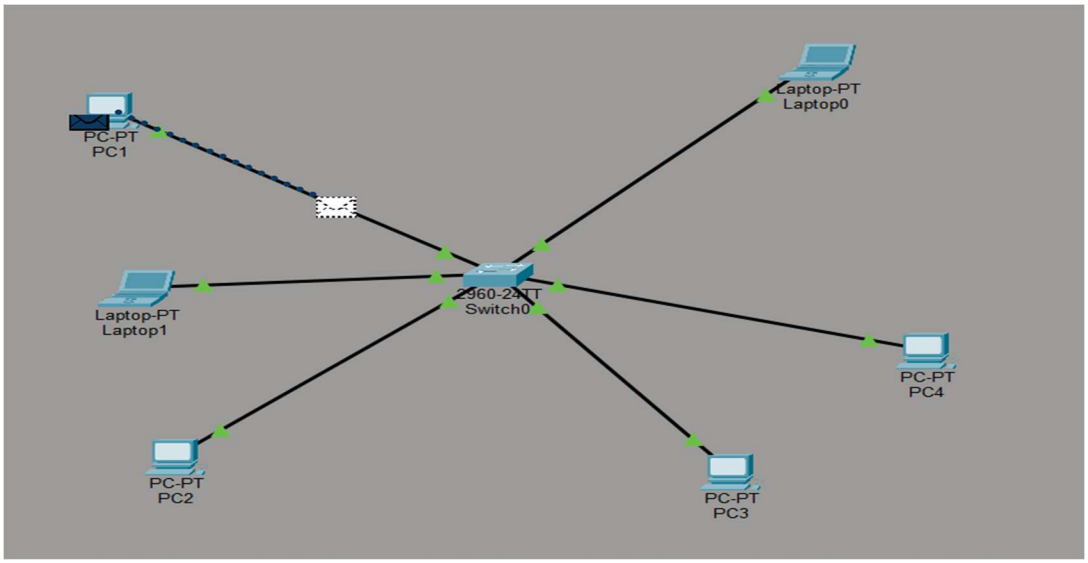
PC1 – 192.168.1.1 / 255.255.255.0
PC2 – 192.168.1.2 / 255.255.255.0
PC3 – 192.168.1.3 / 255.255.255.0
PC4 – 192.168.1.4 / 255.255.255.0
Laptop0 – 192.168.1.5 / 255.255.255.0
Laptop1 – 192.168.1.6 / 255.255.255.0

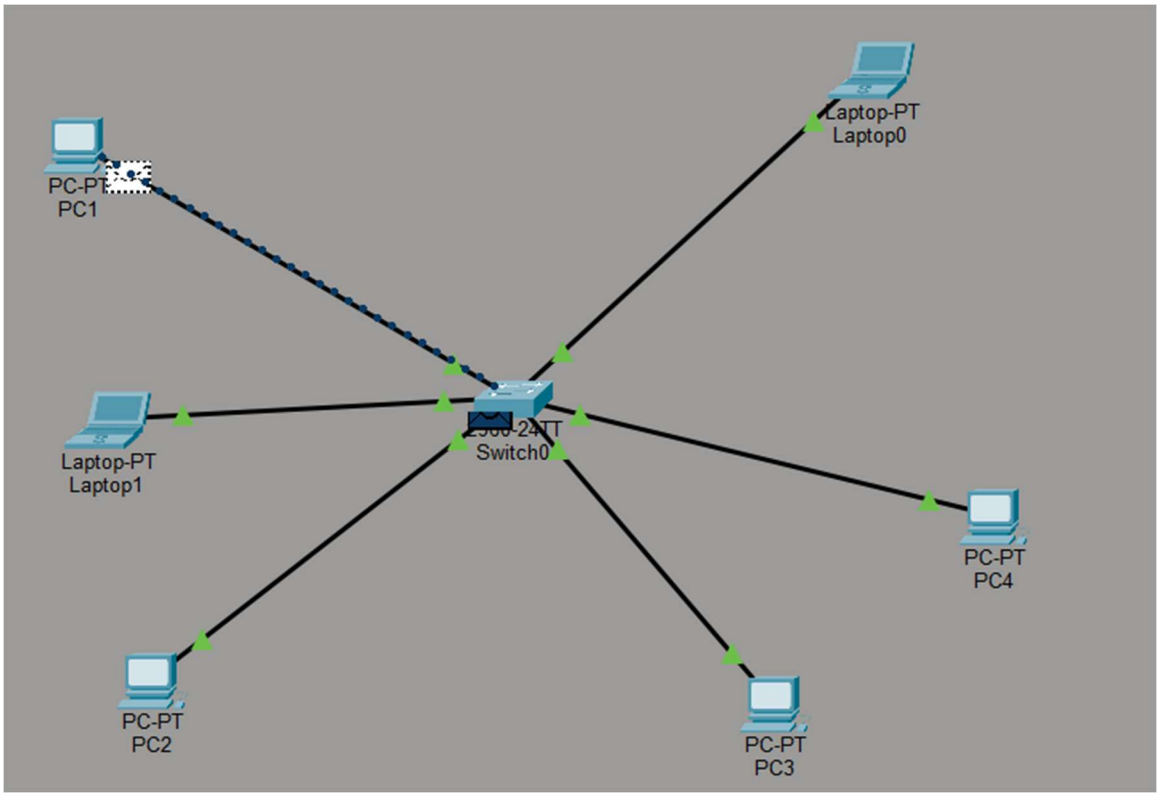
OBSERVATIONS

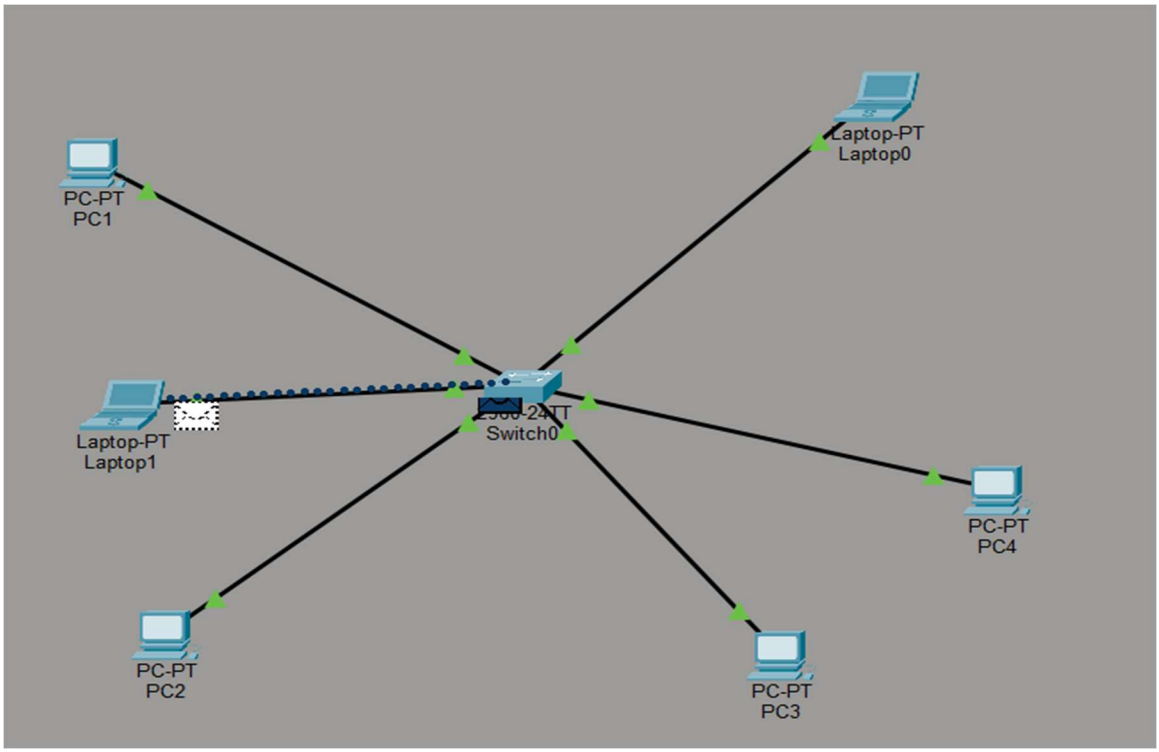
- All devices were successfully connected.
- Ping results showed 0% packet loss.
- TTL value observed was 128.
- Round trip time observed was 0–4 ms.
- Simulation mode showed proper packet forwarding through Switch0.

RESULT

All three topologies were successfully implemented.
Packets Sent = 4
Packets Received = 4
Packet Loss = 0%
Minimum Time = 0 ms
Maximum Time = 4 ms
TTL = 128




Laptop-PT
Laptop0
PC-PT
PC1
Laptop-PT
Laptop1
2960-24TT
Switch0
PC-PT
PC4
PC-PT
PC2
PC-PT
PC3


Laptop-PT
Laptop0
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PC1
Laptop-PT
Laptop1
2960-24TT
Switch0
PC-PT
PC4
PC-PT
PC2
PC-PT
PC3

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=4ms TTL=128
```

Event List

Vis.	Time(sec)	Last Device	At Device
	1.724	--	Switch0
	1.725	Switch0	PC4
	1.725	Switch0	PC1
	1.725	Switch0	PC3
	1.725	Switch0	PC2
	1.725	Switch0	Laptop0
	1.725	Switch0	Laptop1
	1.725	--	PC1
	1.726	PC1	Switch0
	1.727	Switch0	Laptop1
	1.728	Laptop1	Switch0
	1.729	Switch0	PC1
	2.730	--	PC1
	2.731	PC1	Switch0
Visible	2.732	Switch0	Laptop1

CONCLUSION

Star topology provided stable and efficient communication with zero packet loss.
Mesh topology provided high reliability due to multiple paths.
Bus topology was economical but less reliable compared to Star and Mesh.
Based on the experiment, Star topology is most practical for modern networks.